

Climate Shocks, Health Finance, and Economic Outcomes

County-Level Evidence from the United States, 2011–2023

Thesis Committee Update · March 2026

State-Level Results

Two-way FE: State + Year, $N \approx 550-650$, cluster = State

State-Level: Key Significant Results

Two-way FE (State + Year), state-clustered SEs; * $p < 0.10$, ** $p < 0.05$,

*** $p < 0.01$

Employer Premiums (\$2025)

Predictor	Coef.	
Extreme Drought ($h=0$)	+63.6	**
Severe Drought ($h=0$)	+86.4	***
Severe Drought (lag 1)	+65.1	**
Pct PM _{2.5} days	+20.8	***
Pct PM ₁₀ days	+22.7	***
Pct Ozone days	+19.7	***

Medical Debt Share (pp)

Predictor	Coef.	
Cold Shock (lag 1)	+0.0135	**

Medical Debt Median (\$2025)

Predictor	Coef.	
High HDD ($h=0$)	+58.9	**
Severe Drought ($h=0$)	-66.8	***
Severe Drought (lag 1)	-37.5	**

Medicare per Enrollee (\$)

Predictor	Coef.	
High CDD (lag 1)	+144.7	**
High CDD (lag 2)	+137.2	***
High HDD (lag 1)	-110.1	*
High HDD (lag 2)	-192.7	***

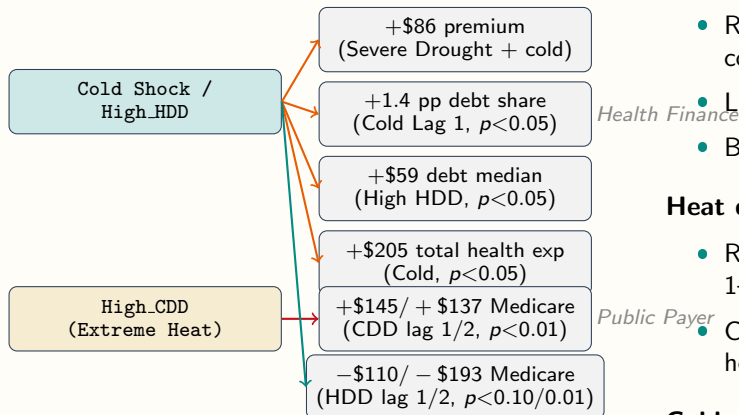
Cold Shock → Total Health Exp: +\$205.3**

Severe Drought → Medicaid/Enrollee: -\$524.5***

State-Level: Cold and Heat Have Opposite Channels

State-level FE; March

2026



Cold channel:

- Raises premiums via drought co-occurrence
- Lagged debt accumulation
- Boosts total system expenditure

Heat channel (CDD):

- Raises Medicare spending with 1–2 year lag
- Consistent with elderly heat-related utilisation

Cold channel (HDD):

- Reduces Medicare with 1–2 year

County-Level Results

Two-way FE: County + Year, $N \approx 23k-34k$, cluster = State

County: Heat Raises Debt Severity; Cold Raises Debt Prevalence

Outcomes: Medical Debt Share (pp) and Median Debt (\$2023); state-clustered SEs

Z_Temp (Relative Heat)

	UW	PW	
<i>Debt Share (pp)</i>			
<i>h=0</i>	-0.0025	-0.0000	
<i>Lag 2</i>	-0.0027**	-0.0014*	UW =
<i>Debt Median (\$)</i>			
<i>h=0</i>	+14.9*	+18.5***	
<i>Lag 2</i>	+19.5**	+16.2***	

unweighted; PW = pop-weighted.

Pattern: heat reduces share (protective?) but raises median severity — composition effect.

High_HDD (Extreme Cold Burden)

	UW	PW
<i>Debt Share (pp)</i>		
<i>h=0</i>	+0.0038**	+0.0054**
<i>Lag 2</i>	+0.0031*	+0.0037*
<i>Debt Median (\$)</i>		
<i>h=0</i>	+10.6	+31.2**

Controls (both outcomes):

Uninsured Rate +*** (large)
HH Income —**

County: Cold Dominates Premium Signal; Heat Effect is Lagged

Outcome: Benchmark Silver Premium (\$2023 real); county FE + Year

FE

High_HDD — Cold Burden

Spec	UW	PW	Cluster
$h=0$	+28.1***	+21.9**	State
$h=0$	+28.1***	+21.9***	RA
Lag 1	-24.5*	-20.8	State

High_CDD — Heat Burden (lag)

Spec	UW	PW	Cluster
Lag 1	+8.7	+31.4***	RA
Lag 2	+21.6*	+35.3***	RA

Z_Temp — Relative Temperature

Spec	UW	PW	Cluster
$h=0$	-10.9**	+2.7	RA
Lag 1	+3.8	+5.4**	RA
Lag 2	+1.0	+4.2**	RA

- Cold burden contemporaneously raises premiums across all specs
- Heat reduces premiums at $h=0$ but raises them with 1–2 year lag
- RA-clustering tightens SEs; results robust
- Drought (PDSI lag 1): +\$3.9 in RA spec ($p=0.05$)

County: Heat Lag Reduces Hospital Bad Debt; Raises HH Income Anomaly

Outcomes: Hosp. Bad Debt per Capita (\$) and Median HH Income (\$2023)

Hospital Bad Debt / Capita

Predictor	UW	PW
Z_Temp Lag 2	-2.41***	-2.08***
High_HDD ($h=0$)	+5.33***	+4.23**
PDSI Lag 1	+0.57*	—
Uninsured	+***	

Heat lag reduces bad debt — possible reduction

in utilisation or elective procedure deferral.

Cold raises bad debt contemporaneously.

Median HH Income (\$2023)

Predictor	UW	PW
Z_Temp ($h=0$)	-71**	-88***
Z_Temp Lag 1	-36	-64***
Z_Temp Lag 2	-71**	-60**
High_CDD ($h=0$)	-247**	-3
High_HDD ($h=0$)	+161**	+173***

Relative heat consistently **reduces** HH income.

Cold (HDD) **raises** income — possible energy/agriculture mix.

County: Drought Depresses Income; Cold Suppresses Employment

Outcomes: Per Capita Personal Income (\$) and Civilian Employment (persons)

Per Capita Personal Income

Predictor	UW	PW
PDSI Lag 2	-8.5	-112.9**
High_CDD Lag 1	+480**	-116
<i>RA-level robustness (pop-wtd):</i>		
PDSI ($h=0$)		-205.7***
PDSI Lag 1	-54.5	

Strong drought signal at Rating-Area level:

-\$206 per capita per PDSI unit ($p<0.001$).

Civilian Employment (persons)

Predictor	UW	PW
Z_Temp Lag 2	+788**	+3216
High_HDD ($h=0$)	-468**	+2587
High_HDD Lag 2	-678**	+3988
Z_Precip ($h=0$)	+46	-5742***
Z_Precip Lag 1	-88	-3803***

Pop-weighted spec: dry conditions strongly reduce employment in populated counties. Unweighted: cold suppresses employment, heat lag raises it.

Cross-Level Synthesis

What do state and county results agree on?

State and County Pipelines Tell a Consistent Story

Key findings replicated across both levels of analysis

Finding	State	County
Cold / High HDD → higher health finance costs	✓	✓
Drought → reduced Medicaid / Medicare spending	✓	✓ (PDSI)
Heat → household income loss	—	✓ (Z_Temp***)
HDD → medical debt (share & median)	✓ (Median)	✓ (Share+Med)
Poor air quality → higher premiums	✓ (PM _{2.5} /O ₃ ***)	not modeled
Drought / dry conditions → employment loss	—	✓ (Z_Precip***)

Model Diagnostics and Robustness

Both pipelines pass standard specification checks

State-Level

- $N = 550\text{--}650$;
Within- $R^2 = 0.09\text{--}0.21$
- Pct_N02 dropped by fixest (collinear with other AQI vars)
- VIF max ≈ 5.3 after PDSI-only drought block
- CDD/HDD thresholds: national p80, 1990–2000 baseline (corrected from within-state all-years)
- `is_high_hdd` newly added this session

County-Level

- $N = 23\text{k--}34\text{k}$; Within- R^2 up to 0.81 (HH Income), 0.44 (PCPI)
- Rating-Area clustered SEs: tighten confidence intervals; core results survive
- RA-level robustness aggregation: pop-weighted, lags re-estimated
- **Debt exclusion:** CO 2023 removed (HB23-1126, Aug snapshot); 60 obs excluded
- Z-scores anchored to 1990–2000 per-county baseline (no look-ahead)

Next Steps

Planned extensions to the empirical framework

Empirical Extensions

- **Event study / local projections:** IRFs for High_CDD, High_HDD, Is_Extreme_Drought, High_AQI_Max — dynamic impulse-response over $h = 0-4$
- **County unemployment:** BLS LAUS integration planned (currently proxied by ACS employment)
- **Mechanism decomposition:** separate cold-weather respiratory illness channel from energy-poverty channel

Writing and Defence

- **Dissertation chapter draft:** integrate state and county findings into unified narrative
- **Heterogeneity analysis:** rural vs. urban counties; high vs. low baseline AQI
- **Policy counterfactuals:** ACA Medicaid expansion interaction with climate shocks

All pipeline code version-controlled;
analysis-ready datasets documented in
CLAUDE.md.